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# =====
# Image Sharpening using High-Boost Filtering
# Google Colab
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Apply Gaussian Blur
blurred = cv2.GaussianBlur(img_rgb, (9, 9), 2)

# High-Boost Factor
A = 1.5

# Convert images to float
img_float = img_rgb.astype(np.float32)
blurred_float = blurred.astype(np.float32)

# High-Boost Filtering
high_boost = A * img_float - blurred_float

# Clip values to valid range
high_boost = np.clip(high_boost, 0, 255).astype(np.uint8)

# Display Images
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(blurred)
plt.title("Blurred Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(high_boost)
plt.title("High-Boost Sharpened Image")
plt.axis("off")

plt.show()

# Save Output
cv2.imwrite("High_Boost_Output.jpg",
            cv2.cvtColor(high_boost, cv2.COLOR_RGB2BGR))

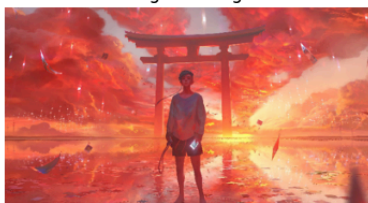
print("Output saved as High_Boost_Output.jpg")

# Download output
files.download("High_Boost_Output.jpg")
```

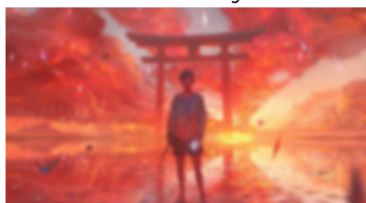
[Choose Files](#) Screenshot... 143039.png

Screenshot 2026-07-20 143039.png(image/png) - 169750 bytes, last modified: 20/7/2026 - 100% done
 Saving Screenshot 2026-07-20 143039.png to Screenshot 2026-07-20 143039.png

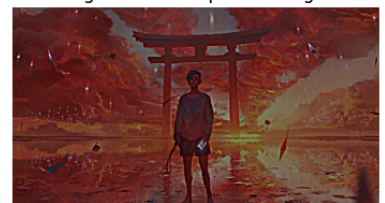
Original Image



Blurred Image



High-Boost Sharpened Image



Output saved as High_Boost_Output.jpg

